

25AM203 ARTIFICIAL INTELLIGENCE

Hours Per Week :

L	T	P	SL	C
2	2	2	2	4

PREREQUISITE KNOWLEDGE: Probability & statistics.

COURSE DESCRIPTION AND OBJECTIVES:

The primary objective of this course is to introduce the basic principles, techniques, and applications of Artificial Intelligence. In addition to this, student will understand the building blocks of AI such as search, knowledge representation, inference, logic and learning. This course enables the students to develop a small AI system for real time problems.

MODULE-1

UNIT-1

12L+12T+12P+12SL=48 Hours

INTELLIGENT SYSTEMS:

Introduction, what is AI, Examples of AI systems, Brief history of AI Agent, Agents and environments, Structure of agents, the concept of rationality, the nature of environments, Types of agents, problem solving approaches to typical AI problem.

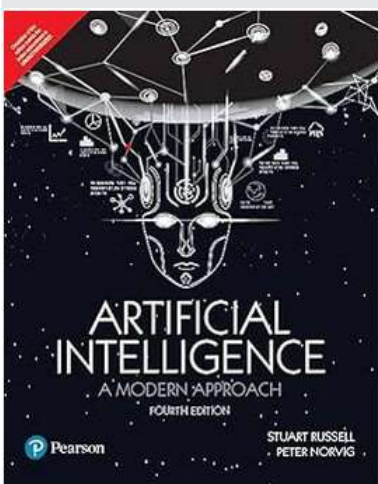
UNIT-2

PROBLEM SOLVING:

State Space Problem; Searching: Uniform search, Informed Search: Solving problems by searching: Examples, Basic Search Strategies, Iterative Deepening DFS, Best First Search, Heuristic functions, A* algorithm, AO* algorithm.

PRACTICES:

- In the classical vacuum cleaner problem, we have two rooms and one vacuum cleaner. There is dirt in both the rooms and it is to be cleaned. The vacuum cleaner is present in any one of these rooms. Find the solution, how we can reach to reach a state in which both the rooms are clean and are dust free.
- In this problem, three missionaries and three cannibals must cross a river using a boat which can carry at most two people, under the constraint that, for both banks, that the missionaries present on the bank cannot be outnumbered by cannibals. The boat cannot cross the river by itself with no people on board. Find the solution, how to solve the given problem.
- You are given two jugs, a 4-gallon one and a 3-gallon one, a pump which has unlimited water which you can use to fill the jug, and the ground on which water may be poured. Neither jug has any measuring markings on it. Find the solution, how can you get exactly 2 gallons of water in the 4-gallon jug?
- There is a farmer who wishes to cross a river but he is not alone. He also has a goat, a wolf, and a cabbage along with him. There is only one boat available which can support the farmer and either of the goat, wolf or the cabbage. So at a time, the boat can have only two objects (farmer and one other). But the problem is, if the goat and wolf are left alone (either in the boat or onshore), the wolf will eat the goat. Similarly, if the Goat and cabbage are left alone, then goat will eat the cabbage. The farmer wants to cross the river with all three of his belongings: goat, wolf, and cabbage. What strategy he should use to do so?
- Either place a block that doesn't have other blocks stacked on top of it on another block with the same behavior, or on the table. The initial and the goal state are described by the exact position of each block. Find the solution, how to solve the given problem.
- Given a 3×3 board with 8 tiles (every tile has one number from 1 to 8) and one empty space. The objective is to place the numbers on tiles to match the final configuration using the empty



- <https://www.amazon.in/Artificial-Intelligence-Modern-Approach-4e/dp/9356063575>

space. We can slide four adjacent (left, right, above, and below) tiles into the empty space. Find the solution, how to solve the given problem by using A* search algorithm.

- The rules of tic-tac-toe on the 3×3 field are as follows. Before the first turn all the field cells are empty. The two players take turns placing their signs into empty cells (the first player places Xs, the second player places Os). The player who places Xs goes first, another one goes second. Find the solution, how to solve the given problem where the winner is the player who first gets three of his signs in a row next to each other (horizontal, vertical or diagonal).
- Write a Program to Implement Breadth First Search using Python.
- Write a Program to Implement Depth First Search using Python.
- Write a Program to Implement Alpha-Beta Pruning using Python

MODULE-2

UNIT-1

6L+6T+6P+6SL= 24 Hours

LOCAL SEARCH:

Satisfaction Vs Optimization, The Example of N-Queens, Hill Climbing, Drawbacks of Hill Climbing, Hill Climbing with Random Walk & Random Restart, Hill Climbing with Simulated Annealing, Genetic Algorithms.

ADVERSARIAL SEARCH AND ETHICS OF AI: Minimax Algorithm for two player games, An Example of Minimax Search, Alpha Beta Pruning, Analysis of Alpha Beta Pruning. Searching game trees.

ETHICS OF AI: Robustness and Transparency of AI systems, Data Bias and Fairness of AI systems, Accountability, privacy and Human-AI interaction.

UNIT-2

KNOWLEDGE REPRESENTATION & PLANNING:

Propositional logic: Inference in propositional logic, Resolution, Forward chaining, Backward chaining, First order logic: Reasoning patterns in First order logic, Resolution, Forward chaining, Backward chaining, and the planning problem: Planning with state space search, Partial order planning, planning graphs.

UNIT-3

LEARNING AND EXPERT SYSTEMS:

Forms of learning: Supervised Learning, Unsupervised learning, Reinforcement learning, Learning Decision Trees, Ensemble Learning.

Expert Systems - Architecture of expert systems, Roles of expert systems - Knowledge Acquisition Meta knowledge, Heuristics. Typical expert systems - MYCIN, DART, XOON, Expert systems shells

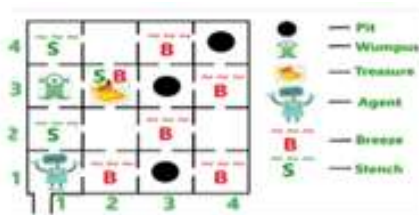
PRACTICES:

- With logic programming, compare expressions and find out unknown values.
- The Wumpus world is a cave with 16 rooms (4×4). Each room is connected to others through walkways (no rooms are connected diagonally). The knowledge-based agent starts from Room [1, 1]. The cave has – some pits, a treasure and a beast named Wumpus. The Wumpus cannot move but eats the one who enters its room. If the agent enters the pit, it gets stuck there. The goal of the agent is to take the treasure and come out of the cave. The agent is rewarded, when the goal conditions are met. The agent is penalized, when it falls into a pit or being eaten by the Wumpus. Some elements support the agent to explore the cave, like -The Wumpus's adjacent rooms are stench. -The agent is given one arrow which it can use to kill the Wumpus when facing it (Wumpus screams when it is killed). – The adjacent rooms of the room with pits are filled with breeze. -The treasure room is always glittery. Find the Wumpus presented room.
- You are on one side of a river with a wolf, a goat, and a cabbage. You want to transport all three to the other side of the river, but you can only transport one object at a time. You cannot leave the wolf and the goat alone, or the cabbage and the goat alone; you are the only thing keeping them from eating each other. How can you transport everything from one side of the river to the other? Formulate it in terms of a Planning Domain Definition Language (PDDL).
- In crypt arithmetic problem, the digits (0-9) get substituted by some possible alphabets or

SKILLS:

- ✓ Know various AI search algorithms (uninformed, informed, heuristic, constraint satisfaction, genetic algorithms).
- ✓ Ability to break down complex problems and design AI-driven solutions.
- ✓ Logical reasoning and algorithmic thinking to improve AI models.
- ✓ Know how to build simple knowledge-based systems.
- ✓ Ability to apply knowledge representation, reasoning, and machine learning techniques to real-world problems.
- ✓ An ability to use current techniques, skills, and tools necessary for computing practice
- ✓ Understanding AI ethics, bias in AI, and responsible AI development.

symbols. The task in crypt arithmetic problem is to substitute each digit with an alphabet to get the result arithmetically correct. Find the solution, how to solve the given problem, where we can perform all the arithmetic operations on a given crypt arithmetic problem.



- Write a Program to Implement Travelling Salesman Problem using Python.
- Write a Program to Implement Tower of Hanoi using Python.
- Write a Program to Implement Monkey Banana Problem using Python.
- Write a Program to Implement Tic-Tac-Toe game using Python.

COURSE OUTCOMES:

Upon successful completion of this course, students will have the ability to:

CO No.	Course Outcomes	Blooms Level	Mapping with POs
1	Apply AI search Models and Generic Search strategies for problem solving.	Apply	1, 3
2	Inspect and analyze Logic for representing Knowledge and Reasoning of AI systems and Conduct investigation and implement project using AI learning techniques.	Analyze	2
3	Apply and evaluate the searching strategies to achieve the goal for a given situation.	Apply	6
4	Design different learning algorithms for improving the performance of AI systems.	Apply	4

MAPPING OF SDGS AND IKS

Sustainable Development Goals (SDGs)	Indian Knowledge System
SDG 3 (Good Health and Well-being)	Yes
SDG 4 (Quality Education)	
SDG 6 (Clean Water and Sanitation)	
SDG 9 (Industry, Innovation, and Infrastructure)	
SDG 12 (Responsible Consumption and Production)	

MAPPING OF SUSTAINABLE DEVELOPMENT GOALS (SDGS) AND INDIAN KNOWLEDGE SYSTEM (IKS):



TEXT BOOKS:

1. S. Russel and P. Norvig, "Artificial Intelligence – A Modern Approach", Fourth Edition, Pearson Education, 2010.

REFERENCE BOOKS:

1. Nilsson N.J., Principles of Artificial Intelligence, Springer Verlag, Berlin.
2. Artificial Intelligence Hand book, Vol. 1-2, ISA, Research Triangle Park.
3. Barr A, Fergenbaub E.A. and Cohen PR. Artificial Intelligence, Addison Wesley, Reading
4. David Poole, Alan Mackworth, Randy Goebel," Computational Intelligence: a logical approach",Oxford University Press. 2018.
5. G. Luger, "Artificial Intelligence: Structures and Strategies for complex problem solving", FourthEdition, Pearson Education, 2010.
6. J. Nilsson, "Artificial Intelligence: A new Synthesis", Elsevier Publishers, 2013.
7. Kos Ko B, Neural Networks and Fuzzy system –PHI
8. 5. Haykin S, Artificial Neural Networks-Comprehensive Foundation, Asea, Pearson